

PIGCN: Physics-Inspired Graph Convolution Networks for Heterogeneous Social Event Detection

Yongsheng Yu
Macquarie University
School of Computing
Sydney, NSW, Australia
yongsheng.yu@hdr.mq.edu.au

Congbo Ma
New York University Abu Dhabi
Computer Engineering
Abu Dhabi, United Arab Emirates
cm7196@nyu.edu

Zitai Qiu
Macquarie University
School of Computing
Sydney, NSW, Australia
zitai.qiu@hdr.mq.edu.au

Shan Xue
Macquarie University
School of Computing
Sydney, NSW, Australia
emma.xue@mq.edu.au

Jian Yang*
Beijing Normal-Hong Kong Baptist
University
Faculty of Science and Technology
Zhuhai, PR China
jianyang@bnnu.edu.cn

Jia Wu†
Macquarie University
School of Computing
Sydney, NSW, Australia
jia.wu@mq.edu.au

Abstract

Social event detection (SED) enhances public awareness by clustering large-scale social messages and has been widely applied across diverse domains. While many existing frameworks adopt Graph Convolution Networks (GCN) as backbones and extend them with auxiliary modules to model complex message relationships, they rarely explore the deeper potential of GCN itself. Specifically, the core operation of GCN—smoothed feature aggregation—implicitly assumes that social information diffusion follows a single-pass and independent heat propagation process. However, real-world diffusion is inherently multi-wave and oscillatory, where different messages interact through reinforcement and interference, resembling the principle of wave-particle duality. To address this gap, we propose **Physics-Inspired Graph Convolution Networks (PIGCN)**, a novel model that unifies wave-based propagation and particle-like interactions. Specifically, PIGCN integrates a Helmholtz Graph Filter module to capture spectral wave propagation with oscillatory dynamics, and a Physical Interaction Force mechanism to adaptively adjust edge weights by attracting messages within the same event and repelling those across events. Building on PIGCN, we further incorporate a contextualized text encoder and timestamp encoding to form a comprehensive SED framework. Extensive experiments demonstrate that PIGCN effectively outperforms conventional GCN baselines and achieves superior performance across multiple SED datasets, establishing a new benchmark for balancing architectural simplicity with effectiveness. Our code can be found on GitHub ¹

*Also with Macquarie University, School of Computing, Sydney, NSW, Australia.

†Corresponding author

¹<https://github.com/yuyongsheng1990/PIGCN>



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ACM ISBN 979-8-4007-2307-0/2026/04

<https://doi.org/10.1145/3774904.3792354>

CCS Concepts

• **Information systems** → **Social networks**; **Data mining**; • **Computing methodologies** → **Neural networks**.

Keywords

PIGCN; Helmholtz Graph Filter; Physical Interaction Force

ACM Reference Format:

Yongsheng Yu, Congbo Ma, Zitai Qiu, Shan Xue, Jian Yang, and Jia Wu. 2026. PIGCN: Physics-Inspired Graph Convolution Networks for Heterogeneous Social Event Detection. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792354>

1 Introduction

By 2025, social media had become nearly ubiquitous, with 63.9% of the global population—approximately 5.24 billion users—actively engaging on social platforms like Twitter, Facebook, Instagram, and TikTok ². The exponential growth of user-generated content poses significant challenges for the timely interpretation and understanding of large-scale public discourse. To address this, social event detection (SED) has emerged as a crucial technique for identifying and clustering meaningful real-world events from high-volume and multi-modal social streams. SED has demonstrated broad applications across diverse domains, including disaster risk management [19], financial market analysis [24], recommendation systems [20], and public opinion analysis [21].

A key observation in SED is that social messages inherently form tree-like heterogeneous graphs, comprising diverse node types (e.g., users, hashtags, messages) and multi-relational links (e.g., posting, mentioning, replying). This structural property naturally motivates the adoption of Graph Neural Networks (GNNs), such as GCN [13], GAT [41], HAN [44], and CTN [51], which are well-suited for modeling complex topological dependencies while preserving textual semantics. Compared with traditional textual [2] and topic-based models [3], GNN-based semi-supervised and unsupervised methods have rapidly become the dominant paradigm for SED,

²<https://datareportal.com/reports/digital-2025-global-overview-report>

with representative examples including MVGAN [7], PPGCN [26], FinEvent [27], ETGNN [32], and ReDHAN [49].

Recent SED methods have achieved notable performance by adopting Graph Convolution Networks (GCN) as their backbone and incorporating auxiliary modules to capture complex message relationships [4, 29, 30, 48]. However, these approaches largely overlook the modeling limitations and untapped potential of GCN itself. The GCN core operation—smoothed feature aggregation—implicitly assumes a single-pass, independent heat diffusion process [5, 13], reducing social information flow to a first-order, monotonic propagation.

In contrast, real-world social diffusion is far more dynamic and interactive, involving multiple-wave propagation, oscillation behaviors, and message interactions. Prior studies have revealed these phenomena from different perspectives, including shockwave-like information propagation [53], competitive/cooperative message spreading [52], and diffusion with feedback-driven state transitions [14]. In social event scenarios, incidents such as natural disasters or political crises typically trigger sequential communication waves, ranging from breaking news and public reactions to official clarifications and subsequent discussions. These waves propagate over time as oscillatory diffusion, in which message numbers are either increased by aligned opinions or reduced under competing viewpoints. Through feedback interactions, social topics within the same event may consolidate into dominant narratives or evolve into polarized debates. At the event level, messages associated with the same event tend to attract one another while those from different events repel—exhibiting characteristics of the Wave-Particle Duality.

Such oscillatory and interactive patterns are difficult to capture with conventional first-order heat diffusion, but naturally align with wave-based propagation governed by second-order partial differential equations (PDEs) [50]. This insight motivates us to redesign the graph convolution operator using a second-order PDE and the interaction force mechanism to directly model social propagation dynamics (see Sec. A.2).

To address these challenges, we propose a novel **Physics-Inspired Graph Convolution Networks (PIGCN)**, which simultaneously models global wave-based oscillatory propagation and localized particle-like interaction forces to capture the intrinsic dynamics of real-world social message diffusion. From the wave perspective, PIGCN updates node representations through a Helmholtz-inspired spectral filter module that explicitly models oscillatory, multi-phase propagation over graph structures. This design is grounded in the classical Helmholtz PDE, which governs wave oscillations in physical systems [6]:

$$\frac{\partial^2 u}{\partial t^2} + k^2 u = 0 \xrightarrow{\text{Discretization}} (L + k^2 I)u = 0 \quad (1)$$

where k^2 represents the oscillation rate and L denotes the graph Laplacian operator. This formulation enables spectral node embeddings to encode frequency-aware structural information in graph propagation. From the particle perspective, we further introduce a two-stage physical interaction force mechanism to adaptively reweight edges based on semantic affinity. First, pairwise node distances are mapped into a Gaussian-kernel space to approximate physical forces between messages [36]. The resulting interaction forces are then used to refine existing edge weights, enhancing local structural discrimination by promoting attraction among intra-event

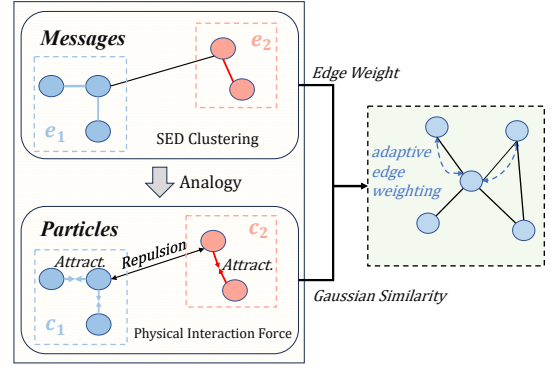


Figure 1: The Physical Interaction Force Mechanism.

Physical Interaction Force models messages as particles that attract or repel via Gaussian-kernel similarity, enabling adaptive edge weighting.

messages while inducing repulsion between inter-event messages (see Fig. 1).

Building upon PIGCN, we further construct a comprehensive social event detection framework by incorporating contextualized text vectors and timestamp encodings. The main contributions of this paper are summarized as follows:

- We are the first to introduce Physics-Inspired Graph Convolution Networks (PIGCN) into the social event detection domain, grounded in a new diffusion perspective that social information propagates through a multi-wave, oscillatory, and interactive process rather than a one-pass, independent flow assumed by standard GCNs. Guided by Wave-Particle Duality, PIGCN jointly captures oscillatory signal transmission (wave perspective) and dynamic message interactions such as attraction and repulsion (particle perspective).
- We derive a novel Helmholtz-based graph filter that integrates oscillation dynamics into the GCN message-passing process via a discretized oscillation rate, enabling more faithful modeling of wave-based propagation patterns in real-world social networks.
- Extensive experiments demonstrate that PIGCN consistently outperforms standard GCNs and achieves superior performance across multiple SED benchmarks, establishing an effective balance between architectural simplicity and modeling power.

2 Preliminaries

Social Event. A *social message* m refers to a media post on social platforms. Messages arriving within a fixed interval Δt (e.g., one day) are grouped into a *message block* $M_t = \{m_i \mid t \leq t_{m_i} \leq t + \Delta t\}$. A *social stream* is a sequence of consecutive message blocks over a time window T_w , defined as $S_t = \{M_t, M_{t+1}, \dots, M_{t+T_w-1}\}$, where S_0 represents the offline setting. A *social event* is defined as a temporally bounded cluster of semantically related messages [12], denoted as $e = \{m_i \mid 1 \leq |m_i| \leq N_{m_e}\}$, where each message belongs to a single event class under the monolingual assumption.

Heterogeneous Social Message Graph. We model the social stream as a Heterogeneous Information Network (HIN) [26, 37], represented by $\mathcal{G} = (\mathcal{V}, \mathcal{R}, \mathcal{T})$. Here, $\mathcal{V}$ contains multiple node types, including messages ($\mathcal{V}_m$), users ($\mathcal{V}_u$), entities ($\mathcal{V}_e$), words ($\mathcal{V}_w$); $\mathcal{R}$ denotes multi-typed relations, encoded by the adjacency matrix $\mathcal{A}$; $\mathcal{T}$ represents temporal information, where message timestamps are scaled into day- and second-components as $\mathcal{T} = \{t_{m_{id}}/10000; t_{m_{is}}/86400 \mid m_i \in S\}$.

Meta-Paths. Meta-paths capture high-order semantic dependencies across heterogeneous node types. A meta-path $p_j \in \mathcal{P}$ is defined as $p_j = \langle v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \dots \xrightarrow{r_k} v_{k+1} \rangle$. We adopt *one-hop meta-paths* to

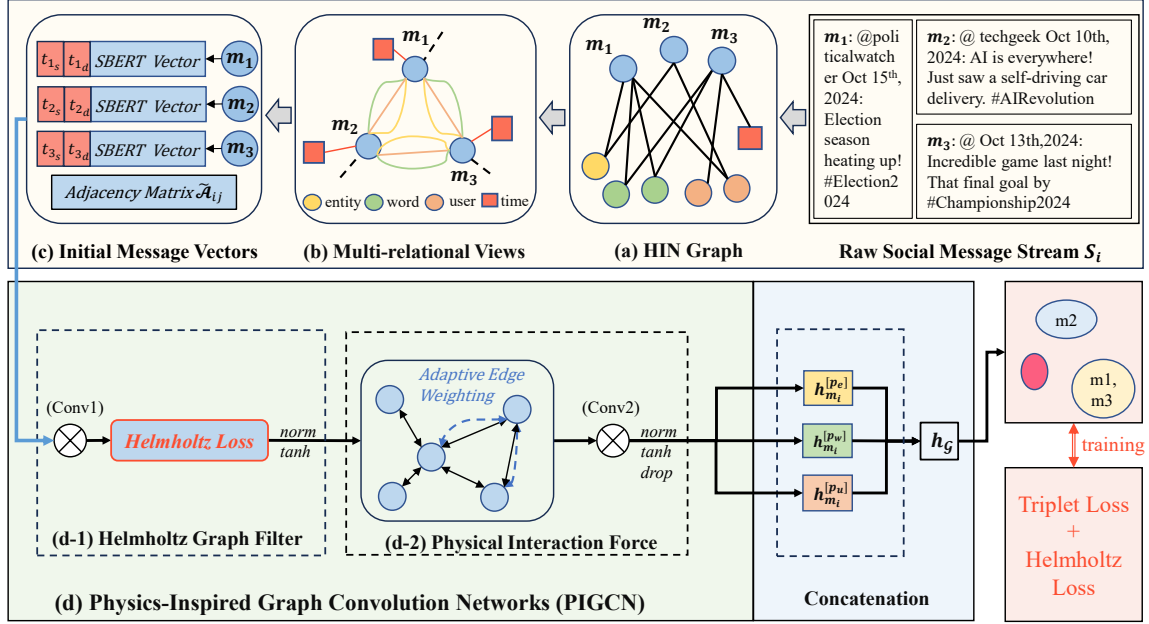


Figure 2: The PIGCN Social Event Detection Framework.

Message features are initialized using SBERT and augmented with temporal encodings to form time-aware vectors. These message vectors are then fed into the (d) Physics-Inspired Graph Convolution Networks (PIGCN) model, which consists of a (d-1) Helmholtz Graph Filter module for oscillatory wave propagation and a (d-2) Physical Interaction Force mechanism for adaptive edge weighting. The resulting embeddings are aggregated across multiple meta-paths to capture heterogeneous relational semantics.

construct indirect message connections via intermediate nodes (e.g., users, entities, or words), such as $p_w = \langle v_{m_i} \rightarrow v_w \rightarrow v_{m_j} \rangle$. These meta-paths decompose the HIN into relation-specific homogeneous subgraphs $\mathcal{G} = (\mathcal{G}_{pu}, \mathcal{G}_{pe}, \mathcal{G}_{pw})$, each encoding a distinct semantic view.

Social Event Detection (SED). Given a social stream S_t , the objective of SED is to partition messages into event clusters:

$$f_t(S_t; \theta_t) = E_t, \quad E_t = \{e_i \mid 1 \leq N_{e_i} \leq |E_t|\} \quad (2)$$

In the offline setting, event detection is initialized as $f_0(S_0; \theta_0) = E_0$, typically solved via K-Means clustering [27]. In the incremental setting, the social streams evolve as $S_t \rightarrow S_{t+1}$ by adding new message blocks and removing outdated blocks. The model detects event clusters on updated streams and is subsequently retrained for future inference [49]:

$$f_t(S_{t+1}; \theta_t) \xrightarrow{\text{detect}} E_{t+1} \xrightarrow{\text{retrain}} f_{t+1}(\theta_{t+1}) \quad (3)$$

3 Methods

This section presents the Physics-Inspired Graph Convolution Network (PIGCN), a unified framework designed to model the multi-wave, oscillatory, and interactive patterns in social information diffusion. As illustrated in Fig. 2, PIGCN operates on a heterogeneous social message graph that is decomposed into multiple weighted homogeneous meta-path graphs to capture diverse semantic relations among messages.

3.1 Initial Message Vector Generation

To construct expressive and time-aware message representations, we initialize message vectors by combining contextualized textual semantics with temporal information.

Specifically, we employ Sentence-BERT (SBERT) model [31] to generate contextualized text vectors for each message $m_i \in S_t$, denoted as $x_{m_i} = \phi_{\text{SBERT}}(m_i)$ (Validated in App. C.3.1). Message timestamps stored in the

OLE_TIME_ZERO format are decomposed into scaled day-level (t_d) and second-level (t_s) components. The initial message representation is then formed by concatenating the semantic and temporal features.

$$h_{m_i}^{(0)} = [x_{m_i}, t_d; t_s] \quad (4)$$

3.2 Physics-Inspired Graph Convolution Networks (PIGCN)

Real-world social diffusion is inherently multi-wave, feedback-driven, and interactive [14, 17, 52, 53], posing challenges for conventional GCNs that rely on smooth, single-pass heat diffusion. To overcome this limitation, PIGCN integrates two complementary physics-inspired mechanisms: (1) a Helmholtz graph filter that models oscillatory and multi-phase wave propagation through a second-order Helmholtz PDE; and (2) a physical interaction force mechanism that adaptively refines edge weights via Gaussian-kernel similarity to simulate attraction–repulsion interactions among messages. By jointly capturing global wave-based propagation and localized interaction forces, PIGCN provides a more faithful and physically grounded representation of real-world social diffusion dynamics.

3.2.1 Wave Perspective: Helmholtz Graph Filter. As shown in Eq. (1), the discretized Helmholtz PDE extends the classic Laplace equation by introducing an oscillation control term k^2 , which selectively favors spectral components approaching the target oscillation rate. To adapt this principle to graph learning, we derive a Helmholtz graph filter (Derivation in App. B.1):

$$g'_\theta(\lambda) = (k^2 + 1) - \lambda \quad (5)$$

where λ denotes the eigenvalues of the graph Laplacian L . Guided by physical intuition, we constrain the practical range of k^2 to mitigate parameter sensitivity in social diffusion tasks. To further balance stability and expressiveness, k^2 is treated as a learnable parameter and initialized to 0.5 (Validated in Sec. 4.5.1), enabling adaptive amplification of signals aligned

with intrinsic oscillatory frequencies while suppressing undesired high-frequency noise.

Based on this filter, the Helmholtz graph convolution operator (HelmholtzGCN) is formulated as:

$$h_{m_i}^{(l+1)} = \sigma \left((k^2 + \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}) h_{m_i}^{(l)} W_h^{(l)} \right) \quad (6)$$

where $\tilde{A}$ denotes the self-loop augmented adjacency matrix, W_h is the linear projection weight matrix in the HelmholtzGCN layer, and σ is the $\tanh$ activation function. We adopt $\tanh$ due to its smooth gradients and non-zero second-order derivatives, which are more suitable for modeling oscillatory dynamics than ReLU (Validated in Sec. 4.5.3).

To further regulate oscillatory behaviors in social propagation, we replace the direct addition of the oscillation term k^2 with a Helmholtz residual loss in the graph convolution, which explicitly enforces the theoretical wave condition:

$$\mathcal{L}_{helm} = \frac{1}{N} \| (k^2 I - \tilde{A}) H_m \|_{Fr}^2 \quad (7)$$

where N is the number of nodes, H_m denotes the hidden feature matrix, and $\| \cdot \|_{Fr}$ represents Frobenius norm. This residual-based regularization integrates seamlessly with standard spectral GCN implementations (e.g., PyG's GCNConv³) while explicitly penalizing deviation from the Helmholtz condition, thereby promoting spectral consistency and preserving long-range signal integrity more effectively than direct parameter injection.

3.2.2 Particle Perspective: Physical Interaction Force. Standard GCNs assume that social messages propagate independently through a single diffusion pass [5, 13], whereas real-world social signals often exhibit mutual reinforcement and interference [52]. In practice, messages associated with the same event tend to cluster—analogueous to attractive forces in physical systems—while messages from different events repel due to semantic divergence. To capture this local semantic coherence, we introduce a two-stage physical interaction force mechanism that adaptively adjusts edge weights based on semantic affinity, enhancing intra-event cohesion while suppressing inter-event interference.

(1) Gaussian Similarity Forces. We first calculate pairwise interaction forces between messages using an exponential Gaussian-kernel similarity:

$$\mathcal{F}_{(m_i, m_j)} = \exp(-\gamma \|h_{m_i} - h_{m_j}\|^2) \quad (8)$$

where γ is a learnable decay rate controlling similarity attenuation (Validated in Sec. 4.5.4). This formulation approximates the exponential distance growth characteristic of tree-structured social graphs while confining computational overhead through mini-batch sampling, thereby maintaining scalability on large-scale graphs.

(2) Edge Weight Refinement. Next, the original edge weights are refined by incorporating the computed physical interaction forces:

$$\hat{W}_{ij}^{phys} = \hat{W}_{ij} + \mathcal{F}_{(m_i, m_j)} \quad (9)$$

where $\hat{W}$ denotes the initial edge weight matrix. In PyG's GCNConv, these refined weights serve as multiplicative coefficients during message passing, directly modulating the contribution of each neighbor to feature aggregation.

After symmetric normalization, the resulting physical interaction force-enhanced graph convolution (PFGCN) updates node representations as:

$$h_{m_i}^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A}^{phys} \tilde{D}^{-\frac{1}{2}} h_{m_i}^{(l)} W_f^{(l)} \right) \quad (10)$$

where W_f denotes the linear projection matrix in the PFGCN layer.

3.2.3 PIGCN Formulation. PIGCN adopts a layer-wise decoupled integration of the Helmholtz Graph Filter and the Physical Interaction Force mechanism within a two-layer architecture. Specifically, the learnable Helmholtz loss is applied in the first GCN layer to regulate frequency-aware propagation, while the Gaussian-based physical force is introduced in the second layer to reinforce local semantic structure. This design explicitly separates frequency-based regularization from structural reinforcement, improving optimization stability and modeling flexibility (Validated in Sec. 4.5.5). The unified PIGCN propagation formulation is defined as:

$$h_{m_i}^{(l+1)} = \sigma \left((k^2 + \tilde{D}^{-1/2} \tilde{A}^{phys} \tilde{D}^{-1/2}) h_{m_i}^{(l)} W_{h/f}^{(l)} \right) \quad (11)$$

where the Helmholtz term models long-range oscillatory propagation, and the physical interaction force promotes local semantic cohesion. Together, these components enhance both the representation quality and propagation fidelity of social message embeddings.

3.3 Optimization Objective

The proposed PIGCN framework is optimized using two complementary loss functions that jointly enhance representation discriminability and spectral stability. The first is the Triplet loss [23], which alleviates class imbalance and structural sparsity in social graphs by enforcing relative similarity constraints:

$$\mathcal{L}_{triplet} = \sum_{(m_i, m_{i+}, m_{i-}) \in T} \max \{ \mathcal{D}(h_{m_i}, h_{m_{i+}}) - \mathcal{D}(h_{m_i}, h_{m_{i-}}) + \delta, 0 \} \quad (12)$$

where $\mathcal{D}(\cdot)$ denotes the Euclidean distance, $m_{i\pm}$ are positive/negative samples relative to anchor m_i , and δ is the margin threshold set to 8 by default. The second is the Helmholtz loss (Sec. 3.2.1), which explicitly regulates spectral behavior and preserves semantic consistency during oscillatory propagation. The overall objective is formulated as:

$$\mathcal{L}_{total} = \mathcal{L}_{triplet} + \mathcal{L}_{helm} \quad (13)$$

where the integration coefficient of the Helmholtz loss is set to 1.0 (Validated in Sec. 4.5.2).

4 Experiments

This section presents the experimental setup and evaluation results for the proposed PIGCN model under both offline and incremental social event detection settings. We conduct extensive experiments across multiple benchmark datasets to compare PIGCN against strong state-of-the-art baselines and to systematically analyze the impact of key design choices and parameter configurations.

Our experiments are guided by the following research questions:

RQ1 How does PIGCN compare with existing state-of-the-art methods in terms of detection performance?

RQ2 How sensitive is HelmholtzGCN to the choice of oscillation rate k^2 ?

RQ3 What is the optimal integration coefficient for the Helmholtz loss?

RQ4 How do different activation functions influence the performance of HelmholtzGCN?

RQ5 Which optimizer is optimal for training HelmholtzGCN?

RQ6 What is the best configuration for the physical interaction force mechanism?

RQ7 What is the most effective integration strategy for combining HelmholtzGCN with the physical force mechanism in PIGCN?

4.1 Experiment Settings

This section describes the experiment setup, including the datasets, baselines, evaluation metrics, and implementation details.

Datasets. We evaluate PIGCN on four widely used social message datasets to assess its generalizability: **Twitter2012** [22], **CrisisLexT** [25],

³https://pytorch-geometric.readthedocs.io/en/latest/_modules/torch_geometric/nn/conv/gcn_conv.html#GCNConv

Table 1: The Offline Social Event Detection Performance (best in red, second-best underlined).

Models	Twitter2012 Dataset				CrisisLexT Dataset			Kawarith Dataset			Twitter2018 Dataset		
	NMI	AMI	ARI	Time (mins)	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI
LDA	40.39	20.60	3.06	0.4	61.02	60.97	65.12	31.94	30.74	5.95	40.78	27.80	3.15
SBERT	86.07	79.18	65.12	0.5	77.68	77.65	84.33	78.40	78.06	72.04	48.76	37.23	12.04
GraphFormer	86.15	79.29	70.24	7.4	<u>89.02</u>	<u>89.00</u>	<u>94.19</u>	92.24	92.12	93.75	68.45	62.11	51.02
FinEvent	90.77	86.05	74.97	<u>3.0</u>	84.26	84.24	91.04	95.29	<u>92.22</u>	96.67	66.02	59.24	28.76
ReDHAN	92.52	88.97	82.51	13.63	87.38	87.36	92.94	<u>92.42</u>	92.30	<u>94.46</u>	73.18	<u>67.45</u>	71.98
HISEvent	88.94	88.40	78.67	13.1	43.87	43.83	37.67	48.02	47.63	45.37	67.48	66.93	<u>51.72</u>
HyperSED	77.27	75.59	54.85	0.5	6.89	6.84	0.84	45.04	44.94	32.71	50.56	48.69	47.47
GCN	92.15	88.34	80.10	5.2	86.39	86.37	92.31	89.44	89.27	91.37	73.41	68.19	44.18
GWNs	91.84	87.88	77.95	6.5	87.89	87.87	93.19	85.55	85.32	84.12	59.54	50.99	21.18
HelmholtzGCN	92.64	89.08	79.50	5.7	87.35	87.34	93.19	88.65	88.47	88.49	68.57	61.60	33.50
PFGCN	<u>92.80</u>	<u>89.35</u>	<u>82.50</u>	4.6	89.36	89.35	94.46	89.83	89.67	89.60	71.98	65.59	46.17
PIGCN	92.93	89.55	80.57	7.0	87.02	87.00	92.94	90.43	90.28	92.47	68.98	62.06	31.50

Kawarith [1], and **Twitter2018** [11] (Detailed in Sec. C.1). Each dataset is split into training, validation, and test sets with ratios of 0.7, 0.2, and 0.1, respectively. By adopting stratified RandomShuffle⁴, we achieve a highly consistent label distribution across splits with Chi-Square significance scores close to 0.99. This strategy ensures both experimental reliability and reproducibility.

Baselines. We compare PIGCN against a diverse of strong baselines in social event detection, including traditional text-based models **LDA** [3], **SBERT** [31], graph-based models **GCN** [13], **GraphFormer** [38], **FinEvent** [27], **ReDHAN** [49], **HISEvent** [4], **HyperSED** [48], and physics-informed approaches **GWNs** [50]. **HelmholtzGCN** and **PFGCN** are two ablated variants of PIGCN: **HelmholtzGCN** retains only the Helmholtz graph filter for oscillatory wave propagation, while **PFGCN** preserves only the Gaussian-based interaction force for adaptive edge weighting.

Evaluation Metrics. Clustering performance is evaluated using three standard metrics: Normalized Mutual Information (NMI), Adjusted Mutual Information (AMI), and Adjusted Rand Index (ARI) [42].

Experiment Configurations. All experiments are conducted on a MacBook Air equipped with an 8-core Apple M3 CPU and 16GB RAM, using Python 3.12.4. The input feature dimension is set to 302, with a hidden dimension of 256 and an output dimension of 762. Models are trained for up to 50 epochs with early stopping (patience=5) and a learning rate of $1e-3$. The margin value of triplet loss is fixed to 8 unless otherwise specified.

4.2 Offline Social Event Detection (RQ1)

We evaluate the effectiveness of the proposed PIGCN model for offline social event detection on four representative datasets. To ensure reliability and reproducibility, each experiment is repeated five times with randomly shuffled data. Table 1 reports the average results, including total runtime (in minutes) with preprocessing.

Across three of the four datasets, PIGCN consistently outperforms both the standard GCN and the state-of-the-art physics-inspired GWN, demonstrating a strong ability to cluster social messages into coherent events. On Twitter2012, PIGCN achieves the best performance across all metrics, reaching an NMI of 92.93%. On CrisisLexT and Kawarith, although PIGCN does not attain the absolute best scores, it still surpasses both GCN and GWN by 1–5%, indicating robust performance in crisis-driven scenarios and strong adaptability to the Arabic-language Kawarith dataset. These results highlight PIGCN’s effectiveness in multilingual and high-noise settings. The underperformance observed on Twitter2018 is primarily attributed to increased graph density and severe class imbalance (33 classes, with

class-14 accounting for 1871 of 4765 samples). In this setting, the Gaussian interaction force disproportionately amplifies majority clusters while overwhelming minority classes, thereby disrupting coherent oscillatory wave propagation. Notably, removing class-14 largely eliminates the performance gap between GCN and PIGCN (84.4% vs. 84.16%, -0.24%). Despite the additional complexity introduced by integrating Helmholtz Graph Filter and the Physical Interaction Force module, PIGCN maintains competitive efficiency, with a runtime of 7 minutes—comparable to GWN’s 6.5 minutes. Overall, PIGCN delivers robust, high-performing, and generalizable results for offline social event detection across diverse datasets and languages.

4.3 Incremental Social Event Detection (RQ1)

In the incremental setting, existing SED pipelines typically follow three stages: (i) Graph update—incorporating new message blocks (e.g., daily data) while removing outdated messages; (ii) Event detection—detecting social events on the updated graph, and (iii) Model maintenance—periodically retraining the SED model to adapt to evolving data. Following this paradigm, we conduct day-level incremental experiments on the Twitter2012 dataset. Starting from an initial stream consisting of two consecutive message blocks, we evaluate three successive social streams (S_1 – S_3). To ensure robustness, each detection process is repeated five times on randomly shuffled data, and the averaged NMI scores—along with total runtime (in minutes, including preprocessing)—are reported in Table 2.

Across all evolving graph snapshots, PIGCN demonstrates stable and competitive performance compared with GCN, achieving NMI scores of 68.22% on S_1 , 61.59% on S_2 , and 69.55% on S_3 . Although GWNs incorporate wave dynamics, they consistently underperform PIGCN on two of the three streams. This performance gap highlights the effectiveness of PIGCN’s unified design, which integrates HelmholtzGCN for modeling global oscillatory propagation and a Physical Interaction Force module for adaptive edge reweighting. Overall, these results validate PIGCN’s suitability for incremental detection, demonstrating strong accuracy, robustness, and efficient adaptability to continuously evolving social graphs.

4.4 Larger Social Event Detection (RQ1)

PIGCN is inherently scalable and introduces no architectural bottlenecks beyond hardware constraints. It supports mini-batch training, leverages neighbor sampling to reduce edge-level computation, and integrates seamlessly with scalable PyTorch Geometric frameworks, making it well-suited for large-scale social graphs. To evaluate scalability, we conduct experiments on enlarged versions of Twitter2012 and CrisisLexT datasets using SBERT vectors (Table 3).

On the enlarged Twitter2012 dataset (10k nodes, 3M edges), PIGCN slightly underperforms the standard GCN in NMI (84.31% vs. 84.57%). In

⁴https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.StratifiedShuffleSplit.html

Table 2: The Incremental SED NMI Performance on Twitter2012 Dataset (best in red, second-best underlined).

Streams	Node	Edges	NMI Metrics and Running Time based on SBERT Vectors							
			GraphFormer	FinEvent	ReDHAN	GCN	GWN	HelmholtzGCN	PFGCN	PIGCN
S ₁	7254	2.4M	65.42	64.42	59.43	65.62	67.22	69.42	67.85	<u>68.22</u>
Time			11.9	10.7	38.0	19.4	21.3	20.9	16.9	19.7
S ₂	7193	2.4M	52.45	54.82	40.56	59.07	61.00	61.34	63.27	<u>61.59</u>
Time			15.8	13.3	34.5	17.8	20.0	22.2	14.9	26.6
S ₃	4777	0.59M	61.43	64.15	51.86	68.39	<u>70.18</u>	69.63	70.65	69.55
Time			6.8	3.8	17.7	10.0	6.7	11.9	9.7	10.6

Table 3: The Larger Social Event Detection under the Offline Setting (best in red, second-best underlined).

Models	Larger Twitter2012: 10k nodes, 3M edges				Larger CrisisLexT: 9k nodes, 30M edges			
	NMI	AMI	ARI	Time (mins)	NMI	AMI	ARI	Time (mins)
GraphFormer	84.55	78.51	50.43	16.0	90.52	90.50	94.38	27.5
FinEvent	83.97	77.69	46.79	16.1	91.23	91.21	94.85	33.2
ReDHAN	<u>84.72</u>	<u>78.83</u>	82.72	65.9	92.27	92.25	95.51	53.8
HISEvent	77.14	76.38	50.30	57.4	41.64	41.56	34.46	245
HyperSED	66.13	63.57	<u>79.61</u>	2.6	51.85	51.83	37.00	2.2
GCN	84.57	78.48	44.72	28.4	91.23	91.21	94.75	29.9
GWN	81.08	73.28	27.77	26.6	90.25	90.23	93.72	54.8
HelmholtzGCN	84.02	77.55	34.29	39.2	93.67	93.66	96.43	27.2
PFGCN	85.51	79.80	43.59	30.3	93.40	93.39	96.30	39.6
PIGCN	84.31	77.93	35.47	33.0	<u>93.49</u>	<u>93.48</u>	<u>96.31</u>	33.7

contrast, on the more complex CrisisLexT dataset (9k nodes, 30M edges), PIGCN achieved the second-highest NMI score (93.49%) with a competitive runtime. This marginal gap reflects the increased difficulty of jointly modeling social oscillatory propagation and interactive dynamics at scale. Consistent with this observation, PFGCN achieves the best performance on the enlarged Twitter2012 dataset, while HelmholtzGCN surpasses other baselines on the larger CrisisLexT when evaluated from the perspective of wave propagation. By consistently outperforming the state-of-the-art physics-informed GWN in dense, high-volume graph settings, PIGCN demonstrates strong robustness, scalability, and effectiveness for large-scale social event detection.

4.5 Parameter Analysis

4.5.1 The Sensitivity Analysis of Oscillation Rate k^2 of HelmholtzGCN (RQ2). A central motivation of PIGCN lies in its physics-inspired design, which inherently reduces hyperparameter sensitivity by grounding key components in well-established physical principles. Among these components, the oscillation rate k^2 plays a pivotal role in HelmholtzGCN, serving as the core physics-informed parameter and a theoretical cornerstone for modeling wave-based social information diffusion.

In physical systems, the parameter k^2 in the Helmholtz PDE governs the balance between diffusion-based smoothing and oscillatory wave propagation [8]. Inspired by this interpretation, we map different k^2 values to analogous social behaviors in message graphs—where $k^2 = 0$ corresponds to standard diffusion, while increasing k^2 introduces oscillatory, feedback-driven propagation. Although this mapping to social diffusion is novel, it is firmly rooted in the classical theory of the Helmholtz equation. The resulting interpretations in social message propagation are summarized as follows:

- $k^2 = 0$: HelmholtzGCN degenerates into standard GCN diffusion, modeling smooth, low-pass propagation without oscillations—analogue to stable information spread without feedback.
- $k^2 = 0.25$: The spectral response remains low-frequency dominant, reinforcing coherent event discussions while suppressing mid-frequency fluctuations.
- $k^2 = 0.5$: Moderate oscillations emerge, activating higher-frequency components and occasional negative responses—corresponding to conflicting opinions or rumor interference (e.g., breaking news followed by public debate and official clarification).
- $k^2 = 1.0$: Stronger oscillatory effects appear, with pronounced mid-to-high frequency activation, resembling polarized debates or rapidly evolving counter-narratives.
- $k^2 = 1.5$ or 2.0 : Excessive oscillations dominate, exhibiting high-frequency positive-negative alternations that resemble electromagnetic signal fluctuations rather than realistic social diffusion.

To validate this physics-inspired social diffusion hypothesis, we conduct a systematic sensitivity analysis of k^2 on HelmholtzGCN using the Twitter2012 dataset with SBERT vectors. Specifically, k^2 is evaluated at fixed values (0.25, 0.5, 1.0, 1.5, 2.0), along with corresponding learnable variants initialized at these values. To further examine which oscillatory regularization is most effective, we implement four HelmholtzGCN variants:

- **HelmholtzGCN_{add}**: directly adds the Helmholtz term to the normalized adjacency matrix (Eq. 6).
- **HelmholtzGCN_{loss}**: applies the Helmholtz regularization loss to both layers (Eq. 7).
- **HelmholtzGCN_{loss_1}**: applies the Helmholtz loss only to the first GCN layer.
- **HelmholtzGCN_{loss_2}**: applies the Helmholtz loss only to the second GCN layer.

These configurations allow us to assess whether oscillatory dynamics are more beneficial for low-level structural propagation, higher-level semantic aggregation, or their combination benefits most from oscillatory dynamics. For reference, the standard GCN baseline ($k^2 = 0$) achieves an NMI of 92.15%. Detailed NMI results and mean runtimes (in minutes, including preprocessing) are reported in Table 4.

Table 4: The Sensitivity Analysis of Oscillation Rate k^2 of HelmholtzGCN.

Models	$k^2 = 0.25$	$k^2 = 0.5$	$k^2 = 1.0$	$k^2 = 1.5$	$k^2 = 2.0$	Times
HelmholtzGCN _{add}						
- fixed k^2	91.34	90.23	90.03	90.36	89.74	6.5
- learnable k^2	91.88	91.87	91.21	90.54	90.11	7.2
HelmholtzGCN _{loss}						
- fixed k^2	92.43	91.48	92.04	92.01	91.45	6.4
- learnable k^2	92.48	92.36	92.01	91.88	92.53	5.8
HelmholtzGCN _{loss_1}						
- fixed k^2	91.75	92.35	91.77	92.09	92.22	5.6
- learnable k^2	92.04	92.64	92.43	92.22	<u>92.51</u>	6.1
HelmholtzGCN _{loss_2}						
- fixed k^2	91.98	92.34	92.22	<u>92.36</u>	92.46	7.1
- learnable k^2	92.26	91.91	<u>92.25</u>	92.60	92.15	6.1

The results reveal that large values of k^2 (e.g., $k^2 = 2.0$) lead to unstable high-frequency oscillations and yield the worst NMI performance (89.74%), confirming their incompatibility with realistic social diffusion. This finding demonstrates that physical intuition effectively narrows the hyperparameter search space by excluding implausible regimes (e.g., $k^2 \geq 1.5$). In contrast, low-pass diffusion ($k^2 = 0$), low-frequency oscillations ($k^2 = 0.25, 0.5$), and moderate mid-frequency dynamics ($k^2 = 1.0$) consistently exceed 90% at the NMI metric, aligning well with real-world social phenomena such as multi-wave discussions, conflicting viewpoints, and counter-narratives.

Across all evaluated settings, HelmholtzGCN variants with learnable k^2 consistently outperform their fixed k^2 counterparts, highlighting the advantage of adaptive oscillation tuning. Moreover, the loss-based implementation of HelmholtzGCN yields a steady 1-2% NMI improvement over the addition-based variant. The best NMI performance (92.64%) is achieved by HelmholtzGCN_{loss_1} with a learnable $k^2 = 0.5$, validating both the effectiveness of loss-based regularization and the physical intuition that moderate oscillatory energy best captures realistic social diffusion dynamics. In summary, the strong performance, stability, and efficiency of the learnable $k^2 = 0.5$ configuration justify adopting HelmholtzGCN_{loss_1} as the default setting in PIGCN.

4.5.2 The Sensitivity Analysis of the Helmholtz Loss Integration Coefficient β (RQ3). Building on our earlier observation that the loss-based implementation of HelmholtzGCN consistently outperforms its direct addition counterpart, we integrate the Helmholtz loss into the overall training objective. Its influence is controlled by a weighting coefficient β , which governs the strength of oscillatory regularization. To examine the sensitivity and effectiveness of this integration, we evaluate HelmholtzGCN_{loss} with learnable $k^2 = 0.5$ across a range of β values (0.1, 0.25, 0.5, 0.75, 1.0).

As illustrated in Fig. 3, when $\beta = 0.1$, HelmholtzGCN_{loss} behaves similarly to a standard GCN, with model behavior dominated by smoothed low-pass propagation and minimal oscillatory regularization—serving as a strong baseline. As β increases, performance steadily improves under the adaptive tuning of k^2 , reaching a peak NMI of 92.36% at $\beta = 1.0$. This trend indicates that stronger Helmholtz regularization, when coupled with learnable oscillatory dynamics, leads to more effective and discriminative embedding learning. Overall, these results confirm that the Helmholtz loss can be fully and stably integrated into the optimization objective under the learnable $k^2 = 0.5$ setting, striking a favorable balance between physics-inspired adaptive regularization and task-specific representation learning.

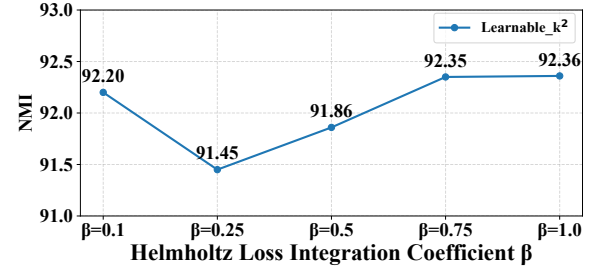


Figure 3: Helmholtz Loss Integration Coefficient β Analysis.

4.5.3 The Exploration of Activation Function of HelmholtzGCN (RQ4). Table 5 compares the impact of different activation functions in the HelmholtzGCN model on the Twitter2012 dataset with SBERT vectors. The analysis aims to identify which nonlinearity best supports oscillation-aware graph learning, especially given the Helmholtz graph filter foundation.

Table 5: Various Activation Function Evaluations in HelmholtzGCN.

Models	Activation	SBERT Vectors on Twitter2012			
		NMI	AMI	ARI	Time (mins)
HelmholtzGCN	ReLU	90.23	85.48	76.11	6.9
	Tanh	91.22	86.76	<u>77.52</u>	6.1
	Softplus	89.22	84.02	76.22	8.4
	tanh(Softplus)	91.01	86.85	81.40	17.7
	SiLU	89.90	85.07	76.81	8.9

Among the evaluated activation functions, Tanh achieves the best overall performance, with an NMI of 91.22%, while maintaining a relatively low training time of 6.1 minutes. These results validate Tanh as the most suited activation for modeling wave-based oscillation propagation, as its smooth gradients and non-zero second-order derivatives align well with the requirements of Helmholtz PDE-based modeling. In contrast, SoftPlus and SiLU perform consistently worse across all metrics. Although Softplus offers smoother gradients than ReLU, it lacks the bounded output of Tanh, which is critical for regulating signal amplitude in an oscillatory setting. The composite activation tanh(Softplus) suffers from substantial computational overhead (17.7 minutes), rendering it impractical for large-scale graph applications despite its expressive nonlinearity.

4.5.4 The Exploration of Optimal Configuration of Physical Interaction Force (RQ6). To evaluate the effectiveness of the physical interaction force mechanism, Table 6 compares two force configurations—Sigmoid-based and Gaussian-kernel-based—on the Twitter2012 dataset using SBERT vectors. For the Gaussian force configuration, we further analyze sensitivity to the decay parameter γ by testing fixed values in the range $[0.1 - 0.9]$ as well as a learnable variant initialized at 0.5. In addition, we investigate whether applying the interaction force to both GCN layers (PFGCN) or to a single layer (PFGCN_1 or PFGCN_2) yields better performance.

As shown in Fig 4, among the fixed- γ settings, PFGCN with a Gaussian kernel at $\gamma = 0.7$ achieves the strongest performance (NMI: 93.02%, AMI: 89.66%, ARI: 83.15%), confirming the effectiveness of Gaussian similarity in capturing fine-grained interaction semantics. However, this advantage does not consistently generalize across configurations: when the force is applied to only one layer (e.g., PFGCN_2), performance degrades and runtime increases (e.g., 7.2 minutes), indicating reduced stability and scalability under fixed parameterization.

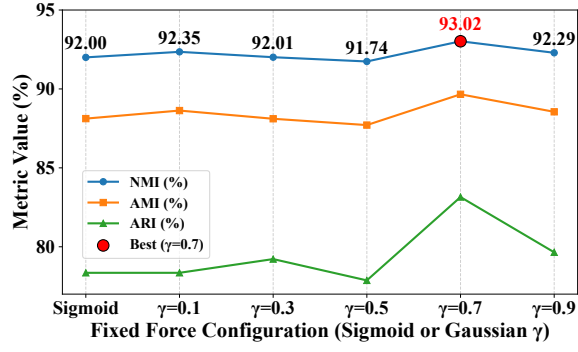


Figure 4: Comparison of Fixed Physical Force Configurations.

Table 6: Exploration of Learnable Physical Interaction Force.

Models	Force Mode	SBERT Text Vectors on Twitter2012			
		NMI	AMI	ARI	Time (mins)
GCN	None	92.15	88.34	80.10	5.2
	Sigmoid	92.00	88.12	78.35	7.1
PFGCN	Gaussian $\gamma=0.7$	93.02	89.66	83.15	5.6
	Gaussian $\gamma_{\text{learnable}}$	92.29	88.57	79.78	5.6
PFGCN ₁	Sigmoid	92.11	88.30	79.77	5.0
	Gaussian $\gamma=0.7$	92.15	88.28	77.86	6.9
	Gaussian $\gamma_{\text{learnable}}$	92.50	88.84	80.51	6.3
PFGCN ₂	Sigmoid	92.56	89.00	80.50	4.8
	Gaussian $\gamma=0.7$	92.68	89.13	81.31	7.2
	Gaussian $\gamma_{\text{learnable}}$	92.80	89.35	82.50	4.6

In contrast, configurations with learnable γ exhibit consistently strong and robust performance across all variants. Notably, PFGCN₂ with learnable γ achieves the second-highest NMI (92.80%) while also attaining the lowest runtime (4.6 minutes), highlighting an efficient and effective trade-off between accuracy and computational cost. These results suggest that applying a learnable Gaussian interaction force exclusively to the second GCN layer enables adaptive, context-aware interaction modeling with minimal overhead.

This behavior reveals an important insight: by the second GCN layer, node embeddings have already incorporated local structural and contextual information from the first layer. Thus, applying the interaction force at this stage operates on semantically enriched representations, leading to more precise interaction modeling. Consistent with this observation, PFGCN₁ with learnable γ —which applies the force directly to raw SBERT vectors—performs comparatively worse (NMI: 92.50%) than its counterpart PFGCN₂ (NMI: 92.80%).

4.5.5 The Evaluation of Integration Strategy of HelmholtzGCN and Physical Interaction Force in PIGCN (RQ7). The Helmholtz loss promotes global oscillatory propagation, whereas the Gaussian interaction force enforces local structural coherence. When these two mechanisms are jointly applied within the same layer, their contrasting objectives can interfere with each other and lead to performance degradation. For instance, applying the Helmholtz loss in both layers while activating the Gaussian force only in the second layer (i.e., Helmloss + Force₂) reduces performance to 91.98% NMI in Table 7, underperforming both the standalone HelmholtzGCN and PFGCN variants. This degradation suggests that simultaneous enforcement

of global oscillatory regularization and local interaction forces may introduce conflicting gradients and semantic misalignment, thereby hindering optimization and convergence.

To address this issue, we adopt a layer-wise decoupling strategy in the two-layer PIGCN architecture. Specifically, the Helmholtz loss with learnable k^2 (initialized at 0.5) is applied in the first layer to guide global wave propagation, while the Gaussian physical force with learnable γ is introduced in the second layer to refine local semantic structure. Additionally, we explore an optional semantic alignment technique—based on cosine similarity regularization [32]—to further harmonize the learning objectives of these two mechanisms. To identify the most effective integration strategy, we evaluate these configurations on the Twitter2012 dataset using SBERT text vectors, with effectiveness and runtime (in minutes, including preprocessing) reported in Table 7.

Table 7: The Integration Strategy Comparison of PIGCN. *Helmloss₁* and *Helmloss* denote the Helmholtz loss applied to the first or both layers, respectively; *Force₂* indicates the Gaussian physical force applied only to the second layer.

Strategy	Configuration	PIGCN Performance			
		NMI	AMI	ARI	Time (mins)
Direct Integration	Helmloss ₁ + Force ₂	92.93	89.55	80.57	7.0
	Helmloss + Force ₂	91.98	88.07	77.97	7.6
Cosine Alignment	Helmloss ₁ + Force ₂	92.24	88.49	79.92	7.8
	Helmloss + Force ₂	92.00	88.11	77.24	8.0

The experimental results demonstrate that direct layer-wise decoupling—applying the Helmholtz loss in the first layer and the Gaussian force in the second—achieves the best overall performance (NMI: 92.93%, AMI: 89.55%, ARI: 80.57%) with a moderate runtime of 7.0 minutes. In contrast, applying the Helmholtz loss to both layers (with Gaussian force only in the second) results in the lowest NMI (91.98%) and incurs additional computational overhead, underscoring the negative impact of semantic conflict. While semantic alignment provides a marginal improvement—raising NMI to approximately 92%—it still falls short of the direct decoupling strategy and introduces extra runtime cost. Overall, these findings indicate that direct layer-wise decoupling, without additional semantic alignment, offers the most effective and efficient strategy for integrating HelmholtzGCN and the physical interaction force within PIGCN.

5 Conclusion

In this work, we propose PIGCN, a novel physics-inspired model for both offline and incremental social event detection that unifies frequency-aware spectral modeling with structurally adaptive message propagation. Specifically, PIGCN integrates a Helmholtz Graph Filter module to capture multi-wave, oscillatory information diffusion and a Physical Interaction Force mechanism to enable adaptive edge weighting with enhanced semantic coherence. Extensive experiments demonstrate that PIGCN consistently outperforms conventional GCN baselines as well as physics-informed GWNs, establishing a new superior benchmark across multiple SED datasets. The proposed model effectively balances architectural simplicity and representational accuracy, while exhibiting only minor scalability limitations on extremely large social graphs. Looking ahead, we plan to extend PIGCN toward online streaming event detection, where lightweight architectures, adaptive graph updates, and real-time inference under limited supervision are critical. Future work will focus on refining the efficiency–accuracy trade-off to support continuous, large-scale social data streams.

Acknowledgments

This work was supported by the Australian Research Council Projects with Nos. DP230100899, LP210301259, DP260100944, and FT250100159.

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A Related Works

A.1 Social Event Detection

Early approaches to Social Event Detection (SED) primarily relied on topic modeling [3] and text embedding methods [2] to cluster semantically similar messages. However, as social platforms evolved into complex heterogeneous networks, these methods proved insufficient for modeling the rich relational and structural dependencies among users, entities, and messages. This limitation motivated the adoption of Graph Neural Networks (GNNs), which have since become the dominant paradigm for SED by jointly modeling textual, structural, and temporal information. Foundational models such as GCN [13], GAT [41], GraphSAGE [9], and HAN [44] laid the groundwork for encoding semantic context and message relations in heterogeneous graphs.

Building on these foundations, subsequent GNN-based SED frameworks addressed challenges including data sparsity, class imbalance, and multilingual heterogeneity. PPGCN [26] mitigates event-level imbalance through contrastive popularity-aware sampling, while ETGNN [32] enhances robustness under weak supervision by integrating evidential learning with Dempster–Shafer theory. FinEvent [27] introduces reinforcement learning to optimize neighborhood sampling, and ReDHAN [49] further enriches structural and semantic modeling via multi-layer residual GATs and dual contrastive objectives.

In parallel, unsupervised SED has attracted increasing attention to reduce reliance on labeled data. HISEvent [4] employs multi-level structural entropy minimization for edge refinement and event clustering, while UHSED [30] leverages hyperbolic embeddings and contrastive learning to cope with label scarcity. MEME-Net [45] combines meta-learning with multi-relational aggregation to support hierarchical event discovery.

Despite their effectiveness, many of these models introduce substantial architectural complexity by incorporating auxiliary mechanisms such as contrastive learning, uncertainty modeling, or knowledge distillation. For example, UCL_{SED} [34] integrates multiple GNN encoders with evidential uncertainty modeling, while CLKD [33] adopts dual knowledge distillation pipelines for low-resource multilingual settings. ADP-SEMEvent [46] reconstructs graph topology using Laplacian similarity and multi-stage structural entropy. Although powerful, such designs often trade efficiency for marginal performance gains.

More recently, research has shifted toward streamlined and principled graph representation learning (GRL) frameworks that balance expressiveness and scalability. GraphHAM [29] models semantic diffusion via automated meta-path encoding in hyperbolic space, while HyperSED [48] improves scalability through semantic anchor graphs and differentiable

hierarchical modeling. These advances reflect a broader trend toward efficient, theoretically grounded GRL architectures tailored for modern SED scenarios.

A.2 PINNs & PIGNNs

Physics-Informed Neural Networks (PINNs), introduced in 2017, embed physical laws—typically expressed as partial differential equations (PDEs)—directly into neural network training. This paradigm has proven effective across diverse physical domains, including wind field prediction [16], power grid optimization [35], and structural crack detection [39]. By incorporating physical priors, PINNs reduce reliance on large labeled datasets and improve robustness to noise and data sparsity, leading to enhanced generalization, data efficiency, and physical consistency.

Motivated by these advantages, recent studies have extended physics-informed learning to graphs, giving rise to Physics-Informed Graph Neural Networks (PIGNNs). These models inject physical priors into message passing to improve generalization and interpretability, particularly on sparse or dynamic graphs. Representative examples include GRAND [5], which formulates GNNs as diffusion-driven continuous time systems; HiD-Net [18], which aligns graph learning with physical diffusion via fidelity constraints; and Graph Wave Networks (GWNs) [50], which employ second-order PDEs and Laplacian operators to model wave propagation with improved spectral stability.

Despite their promise, physics-inspired GNNs remain underexplored, with limited work addressing Social Event Detection (SED)—a domain characterized by dynamic, multi-wave information diffusion and hierarchical social graphs. To bridge this gap, we propose Physics-Inspired Graph Convolution Networks (PIGCN), which embed oscillatory and interaction-aware constraints directly into the GCN operator. Rather than relying on auxiliary objectives or post-hoc optimization, PIGCN enhances structural generalization and event discrimination through principled architectural design, achieving strong performance with competitive computational overhead.

B Methods

B.1 Derivation of the Helmholtz Graph Filter

The HelmholtzGCN is built upon a spectral graph filter that approximates the Helmholtz partial differential equation (PDE) on graphs. This section outlines the derivation by connecting the Helmholtz formulation with the classical spectral GCN framework.

Spectral Graph Convolution. Let $L = U\Lambda U^T$ denote the eigendecomposition of the graph Laplacian, where U contains eigenvectors and $\Lambda = \text{diag}(\lambda_i)$ are the eigenvalues. The graph Fourier transform (GFT) of a signal x is $\hat{x} = U^T x$, with inverse $x = U\hat{x}$. The normalized Laplacian with self-loops is

$$\tilde{L} = I - \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}. \quad (14)$$

By the graph convolution theorem, spectral filtering is expressed as

$$g_G * x = U g_\theta(\Lambda) U^T x = g_\theta(\lambda_i) \hat{x}_i, \quad (15)$$

where $g_\theta(\cdot)$ denotes a learnable spectral filter.

GCN as a Low-Pass Filter. The standard GCN layer [13] can be written as

$$H = \sigma\left((I - \tilde{L})XW\right), \quad (16)$$

which corresponds to the spectral response

$$g_\theta(\lambda) = 1 - \lambda. \quad (17)$$

This formulation acts as a low-pass filter, favoring smooth diffusion by suppressing high-frequency components.

Helmholtz-Inspired Spectral Design. To model oscillatory and multi-wave propagation, we draw inspiration from the Helmholtz equation in the spectral domain:

$$(\lambda + k^2)\hat{x} = 0 \Rightarrow \hat{x}_i \text{ is survived when } \lambda_i + k^2 = 0 \quad (18)$$

This formulation implies that signal components concentrate around eigenvalues satisfying $\lambda_i = -k^2$. Since graph Laplacian eigenvalues satisfy $\lambda_i \geq 0$, this condition cannot be met exactly. Instead, we design a soft spectral filter that emphasizes eigenvalues closer to the target and suppresses those farther away. This leads to a monotonic decay with respect to λ :

$$g'_\theta(\lambda) \propto -(\lambda + k^2). \quad (19)$$

Chebyshev Approximation. For efficiency, we approximate the above behavior using a first-order Chebyshev expansion [10], yielding a linear shift-invariant filter [40]:

$$g'_\theta(\lambda) = C - \lambda. \quad (20)$$

Setting $C = k^2 + 1$ gives the Helmholtz graph filter

$$g'_\theta(\lambda) = (k^2 + 1) - \lambda, \quad (21)$$

with standard GCN as the special case $k^2 = 0$.

HelmholtzGCN Layer. Substituting this filter into the GCN formulation yields

$$\begin{aligned} H_{\text{Helm}} &= \sigma(g'_\theta(\tilde{L})XW) \\ &= \sigma((k^2I + \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2})XW) \end{aligned} \quad (22)$$

which provides a principled and efficient mechanism for incorporating Helmholtz-inspired oscillatory dynamics into graph convolution.

B.2 PIGCN Model Algorithm

The overall procedure of PIGCN is summarized in Algorithm 1.

Algorithm 1: PIGCN Model Algorithm

Input: S_t : social streams; T : timestamps; $\mathcal{A}$: heterogeneous adjacency; $\mathcal{P}$: one-hop meta-paths.

Output: h_G : final message representations.

- 1 **Initialization.** **foreach** $m_i \in S_t$ **do**
- 2 $x_{m_i} \leftarrow f_{\text{SBERT}}(m_i)$;
- 3 $(t_d, t_s) \leftarrow t_{m_i}$;
- 4 $h_{m_i}^{(0)} \leftarrow [x_{m_i}; t_d; t_s]$
- 5 **Graph Construction.** Construct $\mathcal{G}(\mathcal{V}, \mathcal{R}, \mathbb{T})$ from S_t and decompose it into $\{\mathcal{G}_p\}_{p \in \mathcal{P}}$.
- 6 **PIGCN Forward.** $\mathcal{L}_{\text{helm}} \leftarrow 0$;
- 7 **foreach** $\mathcal{G}_p \in \mathcal{P}$ **do**
- 8 Compute hidden features H_m ;
- 9 $\mathcal{L}_{\text{helm}} \leftarrow \frac{1}{N} \|(k^2I - \tilde{A})H_m\|_{F_r}^2$;
- 10 $\mathcal{F}_{m_i, m_j} \leftarrow \exp(-\gamma \|h_{m_i} - h_{m_j}\|^2)$;
- 11 $\hat{\mathcal{W}}^{\text{phys}} \leftarrow \hat{\mathcal{W}} + \mathcal{F}_{m_i, m_j}$;
- 12 Obtain $\tilde{A}^{\text{phys}}$ through multiplication of $\hat{\mathcal{W}}^{\text{phys}}$;
- 13 $h_{m_i}^{(l+1)} \leftarrow \sigma((k^2I + \tilde{D}^{-1/2}\tilde{A}^{\text{phys}}\tilde{D}^{-1/2})h_{m_i}^{(l)}W_{h/f}^{(l)})$;
- 14 **Optimization.** $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{triplet}} + \mathcal{L}_{\text{helm}}$;
- 15 **return** h_G

B.3 Time Complexity Analysis

The overall time complexity of PIGCN is $O(Nd^2 + Rd)$, where N and R denote the numbers of nodes and edges, respectively, and d is the output dimension. The term $O(Nd^2)$ arises from the linear transformations applied to node features in each graph convolution layer, while $O(Rd)$ corresponds to neighborhood aggregation during message passing. Both the HelmholtzGCN and Physical Interaction Force modules operate on sparse adjacency structures, ensuring that computation scales linearly with the number of edges. By leveraging sparse edge_index representations and

optimized scatter-based operations in PyG, PIGCN achieves low practical overhead and near-linear scalability with respect to graph size, making it efficient for large-scale sparse social networks.

C Experiments

C.1 Dataset

- **Twitter2012 Dataset** [22]: A Twitter corpus annotated with real-world events (e.g., protests and political rallies), where each message includes textual content, timestamps, and user-level metadata, enabling temporal and relational modeling of social event propagation.
- **CrisisLexT Dataset** [25]: A large-scale dataset of annotated tweets related to diverse crisis events, designed to study information diffusion and public response during emergencies.
- **Kawarith Dataset** [1]: An Arabic Twitter dataset for crisis event detection, supporting multilingual and cross-lingual analysis of social events in Arabic-language contexts.
- **Twitter2018 Dataset** [11]: A French Twitter dataset from the TwitterNews+ framework, consisting of real-time tweet streams for event detection and topic evolution analysis in French social media.

The statistics of the four SED datasets are summarized in the Table 8.

Table 8: Statistics of the Social Event Detection Datasets

Dataset	#Nodes	#Edges	#Events	#Users
Twitter2012	4762	509 k	57	4355
CrisisLexT	6123	17 M	2	5276
Kawarith	4860	945 k	6	3582
Twitter2018	4765	652 k	33	3063

C.2 Baseline Models

- **LDA** ⁵ [3]: A probabilistic topic model that represents messages as mixtures of latent topics derived from word co-occurrence. While effective for coarse thematic discovery, it ignores contextual semantics and performs poorly on short social texts.
- **SBERT** ⁶ [31]: A transformer-based sentence encoder that produces contextualized embeddings optimized for semantic similarity, serving as a strong text-only baseline for SED clustering.
- **GCN** ⁷ [13]: A classical graph convolution network that aggregates neighborhood information via Laplacian-based smoothing, forming the foundational GNN baseline.
- **GraphFormer** ⁸ [38]: A transformer-based GNN that incorporates self-attention into graph structures to jointly model local interactions and long-range dependencies.
- **FinEvent** ⁹ [27]: A temporal SED framework that uses reinforcement learning to optimize neighborhood sampling and GAT-based encoders to aggregate heterogeneous social signals.
- **ReDHAN** ¹⁰ [49]: A heterogeneous GNN with multi-layer residual attention that captures deep semantic and structural dependencies, enhanced by dual contrastive learning to mitigate structural sparsity and imbalance.

⁵<https://github.com/cemoody/lda2vec>

⁶<https://huggingface.co/sentence-transformers>

⁷<https://github.com/senadkurtisi/pytorch-GCN>

⁸<https://github.com/lucidrains/graph-transformer-pytorch>

⁹<https://github.com/RingBDStack/FinEvent>

¹⁰<https://github.com/yuyongsheng1990/Re-DHAN>

- **HISEvent**¹¹ [4]: A hierarchical SED model that leverages multi-level attention over heterogeneous relations and structural entropy minimization for unsupervised event discovery.
- **HyperSED**¹² [48]: An unsupervised framework that constructs semantic anchor graphs and differentiable partition trees to improve scalability while preserving hierarchical structure.
- **GWNs**¹³ [50]: A physics-inspired GNN that models wave-based signal propagation via second-order PDEs, offering improved spectral expressiveness over diffusion-based graph models.

C.3 Extended Analysis Experiments

C.3.1 Comparison of Initial Text Models. To examine how initial text representations influence semantic quality and SED performance, we conduct comparative experiments on the Twitter2012 dataset using four representative text encoders: Doc2Vec [28], BERT [15], E5 [43], and SBERT [31]. Each encoder is integrated into two fundamental GNN backbones—GCNs and GATs—to assess the interaction between text feature quality and graph-based event detection.

Table 9: Evaluation of Initial Text Models on SED.

Text Model	GCNs (NMI)	GATs (NMI)	Running Time (mins)
Doc2Vec	91.78	89.29	8.2
BERT	89.70	87.46	13.8
E5	91.09	90.73	14.1
SBERT	92.15	90.77	<u>8.6</u>

Table 9 reports the corresponding NMI scores and runtime. SBERT consistently achieves the best performance across both GCNs and GATs, while maintaining competitive efficiency. In contrast, BERT underperforms due to its lack of sentence-level optimization, and E5, although competitive, incurs a higher computational cost. Overall, SBERT provides the most favorable trade-off between semantic expressiveness and efficiency, justifying its adoption as the default text encoder in the PIGCN framework.

C.3.2 PIGCN Robustness on Weaker Text Vectors (RQ1). PIGCN exhibits strong robustness across different initial text features, which is essential for fair and reliable evaluation. Rather than tuning the architecture for peak performance on a single text representation, PIGCN is designed to remain effective under diverse and even weaker semantic features. To further validate this robustness, we evaluate PIGCN using Doc2Vec—a classic shallow text encoder—and compare it with representative baselines on the Twitter2012 dataset.

Table 10: Offline SED on Doc2Vec Vectors

Model	NMI	AMI	ARI	Time (mins)
GraphFormer	91.37	87.21	78.60	4.9
FinEvent	90.09	85.01	75.70	2.7
ReDHAN	92.64	89.13	83.16	15.2
GCN	91.43	87.29	78.45	5.3
GWN	86.96	80.49	67.55	6.0
HelmholtzGCN	91.83	87.89	<u>81.48</u>	5.1
PFGCN	91.74	87.72	77.74	6.9
PIGCN	<u>92.22</u>	<u>88.45</u>	79.47	8.1

As reported in Table 10, PIGCN maintains competitive performance despite the reduced semantic quality of input features. Specifically, PIGCN

achieves an NMI of 92.22%, outperforming both the standard GCN and the physics-inspired GWN model, and remaining close to the strongest based (ReDHAN). These results confirm that PIGCN’s performance gains stem from its physics-inspired propagation and interaction modeling rather than reliance on high-quality text features, demonstrating strong robustness to input feature variations.

C.3.3 Optimizer Exploration on HelmholtzGCN Training (RQ5). HelmholtzGCN is inspired by Physics-Informed Neural Networks (PINNs), where second-order optimization is theoretically appealing due to the smoothness, continuity, and second-order differentiability required by the Helmholtz-based formulation. In contrast, the widely adopted Adam optimizer is a first-order method and does not explicitly exploit curvature information. To examine whether higher-order optimization benefits HelmholtzGCN, we compare Adam with several quasi-second-order optimizers, including AdaHessian [47], Ranger, and L-BFGS, using *tanh* activation throughout.

Table 11: Optimizer Explorations on HelmholtzGCN

Optimizers	Doc2Vec vectors		SBERT vectors	
	NMI	Times	NMI	Times
Adam	91.43	5.0	92.64	5.7
AdaHessian	85.13	6.7	88.67	6.7
Ranger	86.26	5.3	88.10	5.3
L-BFGS	-	> 20	-	> 20

As shown in Table 11, L-BFGS suffers from prohibitively high computational cost (exceeding 20 minutes per run) and is therefore impractical for graph-based training. AdaHessian and Ranger, while designed to incorporate higher-order information, consistently underperform Adam across both Doc2Vec and SBERT settings. These results suggest that, despite the second-order nature of the Helmholtz objective, existing quasi-second-order optimizers do not translate into empirical gains for physics-inspired GNNs. Consequently, we adopt Adam as the default optimizer for HelmholtzGCN, balancing performance and efficiency.

C.3.4 Interpretable Case Study for Practical Application Effectiveness. Interpretable case studies are critical for assessing the practical utility and explainability of SED methods. In our workflow, the message clusters generated by PIGCN are passed to a large language model (LLM) to generate concise, human-readable event summaries. As a case study, we applied this process to the Twitter2012 dataset (referenced in Table 1), selecting the top five clusters, each containing ten representative messages. The resulting summaries are as follows:

- Cluster-5: Termination of Merger Between Aerospace Giants BAE Systems and EADS Due to Political Disagreements.
- Cluster-4: Global Outcry and Support Following the Taliban Attack on Malala Yousafzai.
- Cluster-34: Public Sentiment and Reactions Toward the “Keyshia & Daniel: Family First” TV Show.
- Cluster-37: Turkish Military Intercepts Syrian Passenger Plane Suspected of Transporting Russian Munitions.
- Cluster-9: Chinese Author Mo Yan Wins the 2012 Nobel Prize in Literature.

These coherent and contextually accurate summaries demonstrate PIGCN’s ability to form semantically consistent clusters that closely correspond to real-world events, highlighting both its interpretability and practical effectiveness.

¹¹<https://github.com/SELGroup/HISEvent>

¹²<https://github.com/XiaoyanWork/HyperSED>

¹³<https://github.com/YueAWu/Graph-Wave-Networks>